Geocomputation

Complex Visualisations



Module outline

W1	Reproducible Spatial Analysis	Core Spatial Analysis
W2	Spatial Queries and Geometric Operations	
W3	Point Pattern Analysis	
W4	Spatial Autocorrelation	
W5	Spatial Models	
W6	Raster Data Analysis	Applied Spatial Analysis
W7	Geodemographic Classification	
W8	Accessibility Analysis	
W9	Beyond the Choropleth	Data Visualisation
W10	Complex Visualisations	

This week

- Tidy data and the grammar of graphics.
- Package and code management.
- Working with an agentic assistant.
- Module recap and exam pointers.

Tidy data

Tidy data: structure your data first

- It is often said that 80 per cent of data analysis time goes to cleaning and preparing data (Wickham, 2014).
- Tidy data refers to the structure and organisation of your data set.
- The idea boils down to three principles.
- The principles are brought together in the tidyverse.

Tidy data



Each variable must have its own column

Tidy data



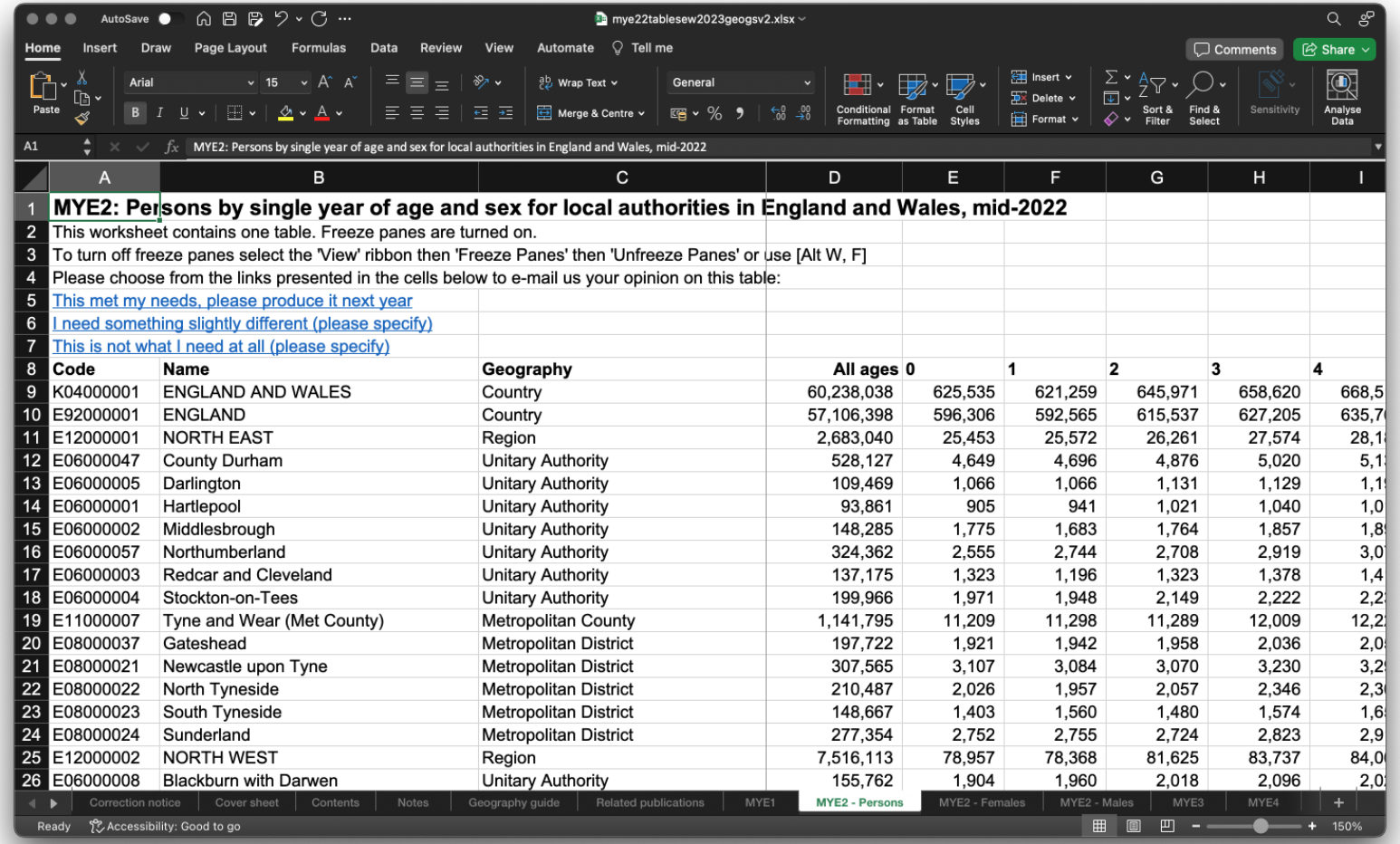
Each observation must have its own row

Tidy data



Each value must have its own cell

Real data are rarely tidy



mye22tablesew2023geogsv2.xlsx

	A	B	C	D	E	F	G	H	I
1	**MYE2: Persons by single year of age and sex for local authorities in England and Wales, mid-2022**								
2	This worksheet contains one table. Freeze panes are turned on.								
3	To turn off freeze panes select the 'View' ribbon then 'Freeze Panes' then 'Unfreeze Panes' or use [Alt W, F]								
4	Please choose from the links presented in the cells below to e-mail us your opinion on this table:								
5	This met my needs, please produce it next year								
6	I need something slightly different (please specify)								
7	This is not what I need at all (please specify)								
8	**Code**	**Name**	**Geography**	**All ages**	**0**	**1**	**2**	**3**	**4**
9	K04000001	ENGLAND AND WALES	Country	60,238,038	625,535	621,259	645,971	658,620	668,5
10	E92000001	ENGLAND	Country	57,106,398	596,306	592,565	615,537	627,205	635,7
11	E12000001	NORTH EAST	Region	2,683,040	25,453	25,572	26,261	27,574	28,1
12	E06000047	County Durham	Unitary Authority	528,127	4,649	4,696	4,876	5,020	5,1
13	E06000005	Darlington	Unitary Authority	109,469	1,066	1,066	1,131	1,129	1,1
14	E06000001	Hartlepool	Unitary Authority	93,861	905	941	1,021	1,040	1,0
15	E06000002	Middlesbrough	Unitary Authority	148,285	1,775	1,683	1,764	1,857	1,8
16	E06000057	Northumberland	Unitary Authority	324,362	2,555	2,744	2,708	2,919	3,0
17	E06000003	Redcar and Cleveland	Unitary Authority	137,175	1,323	1,196	1,323	1,378	1,4
18	E06000004	Stockton-on-Tees	Unitary Authority	199,966	1,971	1,948	2,149	2,222	2,2
19	E11000007	Tyne and Wear (Met County)	Metropolitan County	1,141,795	11,209	11,298	11,289	12,009	12,2
20	E08000037	Gateshead	Metropolitan District	197,722	1,921	1,942	1,958	2,036	2,0
21	E08000021	Newcastle upon Tyne	Metropolitan District	307,565	3,107	3,084	3,070	3,230	3,2
22	E08000022	North Tyneside	Metropolitan District	210,487	2,026	1,957	2,057	2,346	2,3
23	E08000023	South Tyneside	Metropolitan District	148,667	1,403	1,560	1,480	1,574	1,6
24	E08000024	Sunderland	Metropolitan District	277,354	2,752	2,755	2,724	2,823	2,9
25	E12000002	NORTH WEST	Region	7,516,113	78,957	78,368	81,625	83,737	84,0
26	E06000008	Blackburn with Darwen	Unitary Authority	155,762	1,904	1,960	2,018	2,096	2,0

Correction notice | Cover sheet | Contents | Notes | Geography guide | Related publications | MYE1 | MYE2 - Persons | MYE2 - Females | MYE2 - Males | MYE3 | MYE4

Ready | Accessibility: Good to go | 150%

Five common errors in untidy data

- Column headers are values rather than variable names.
- Multiple variables are stored in one column.
- Variables are stored in both rows and columns.
- Multiple observational units are stored in the same column.
- A single observation is stored in multiple tables.

Tidy?

country	year	type	count
Afghanistan	2019	cases	745
Afghanistan	2019	population	19,987,071
Afghanistan	2020	cases	2,666
Afghanistan	2020	population	20,595,360
Brazil	2019	cases	37,737
Brazil	2019	population	172,006,362
Brazil	2020	cases	80,488
Brazil	2020	population	174,504,898
China	2019	cases	212,258
China	2019	population	1,272,915,272
China	2020	cases	213,766
China	2020	population	1,280,428,583

Tidy?

country	year	rate
Afghanistan	2019	745 / 19,987,071
Afghanistan	2020	2,666 / 20,595,360
Brazil	2019	37,737 / 172,006,362
Brazil	2020	80,488 / 174,504,898
China	2019	212,258 / 1,272,915,272
China	2020	213,766 / 1,280,428,583

Tidy?

Cases

country	2019	2020
Afghanistan	745	2,666
Brazil	37,737	80,488
China	212,258	213,766

Population

country	2019	2020
Afghanistan	19,987,071	20,595,360
Brazil	172,006,362	174,504,898
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Tidy?

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Grammar of graphics

Grammar of graphics: a graphic is built in layers

- Graphics are constructed by layering multiple elements of data.
- Values in a dataset serve as aesthetics: attributes that can be visually represented in a graphic.
- Data, scales, coordinate systems, and plot annotations are layered on top of these aesthetics to create the final graphic.

Data, scales and annotations are separate layers

Dataframe

x	*y*	Shape
2	4	a
1	1	a
4	15	b
9	80	b



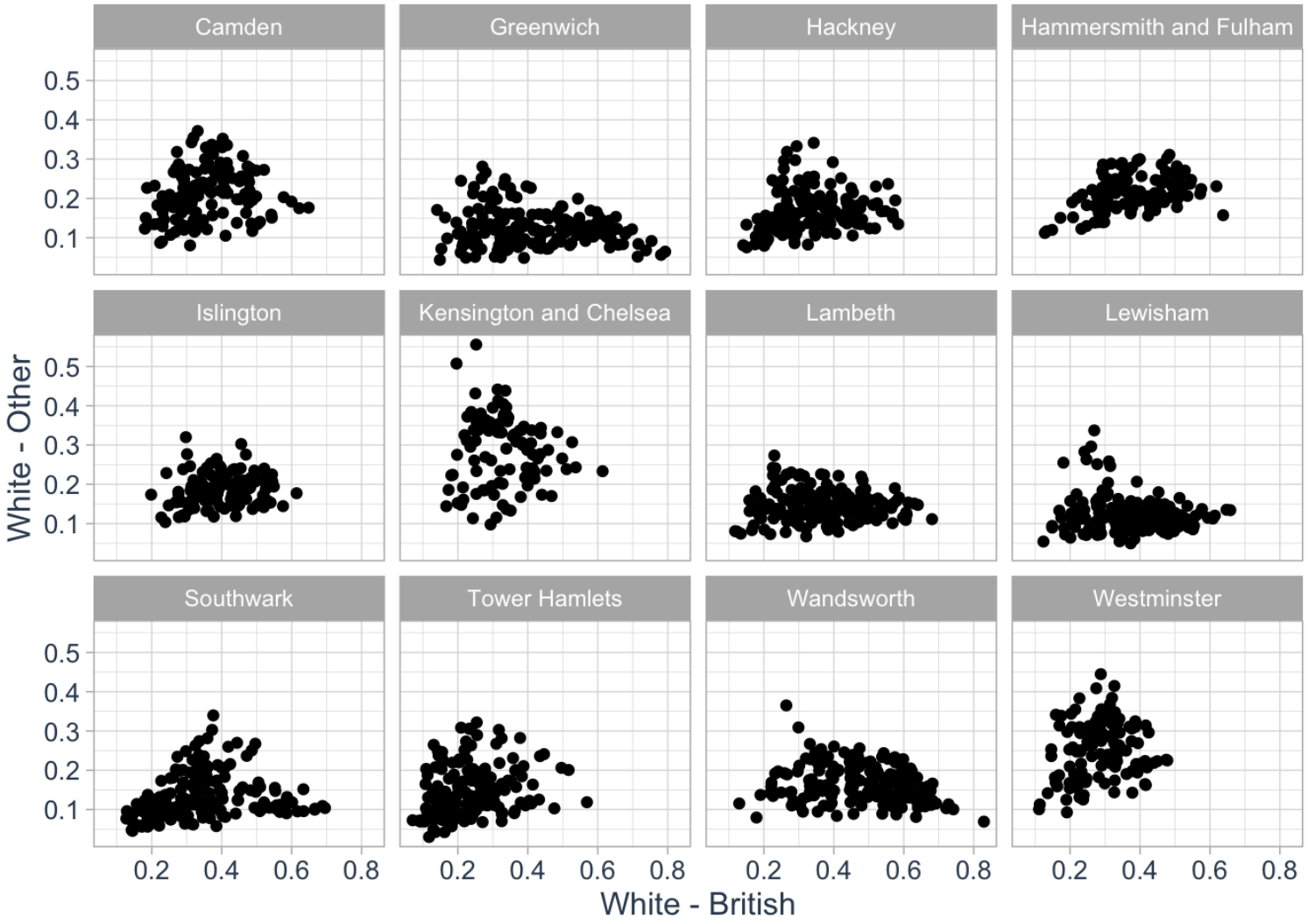
Dataframe values

Dataframe scale

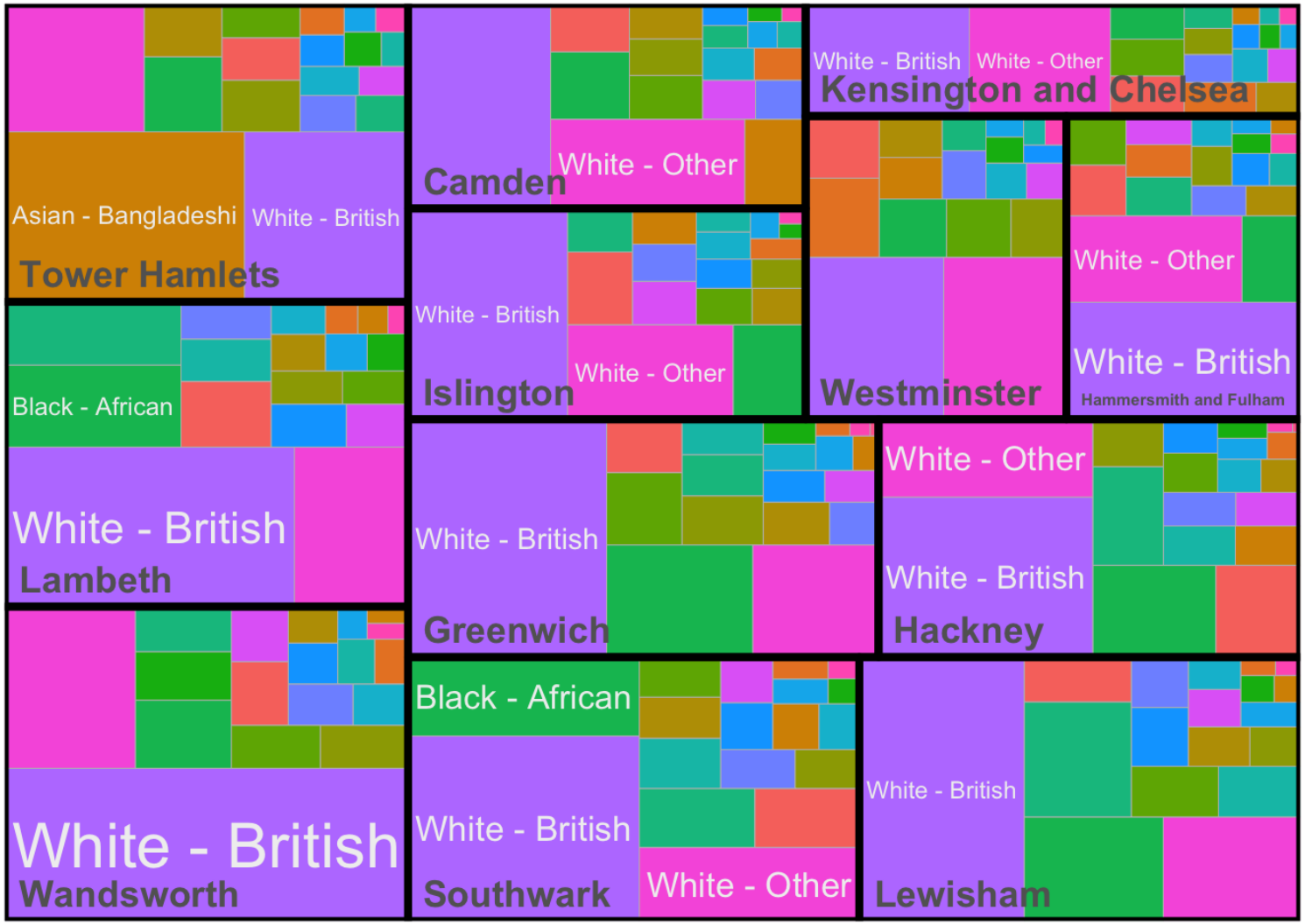
Dataframe annotations

Final Graphic

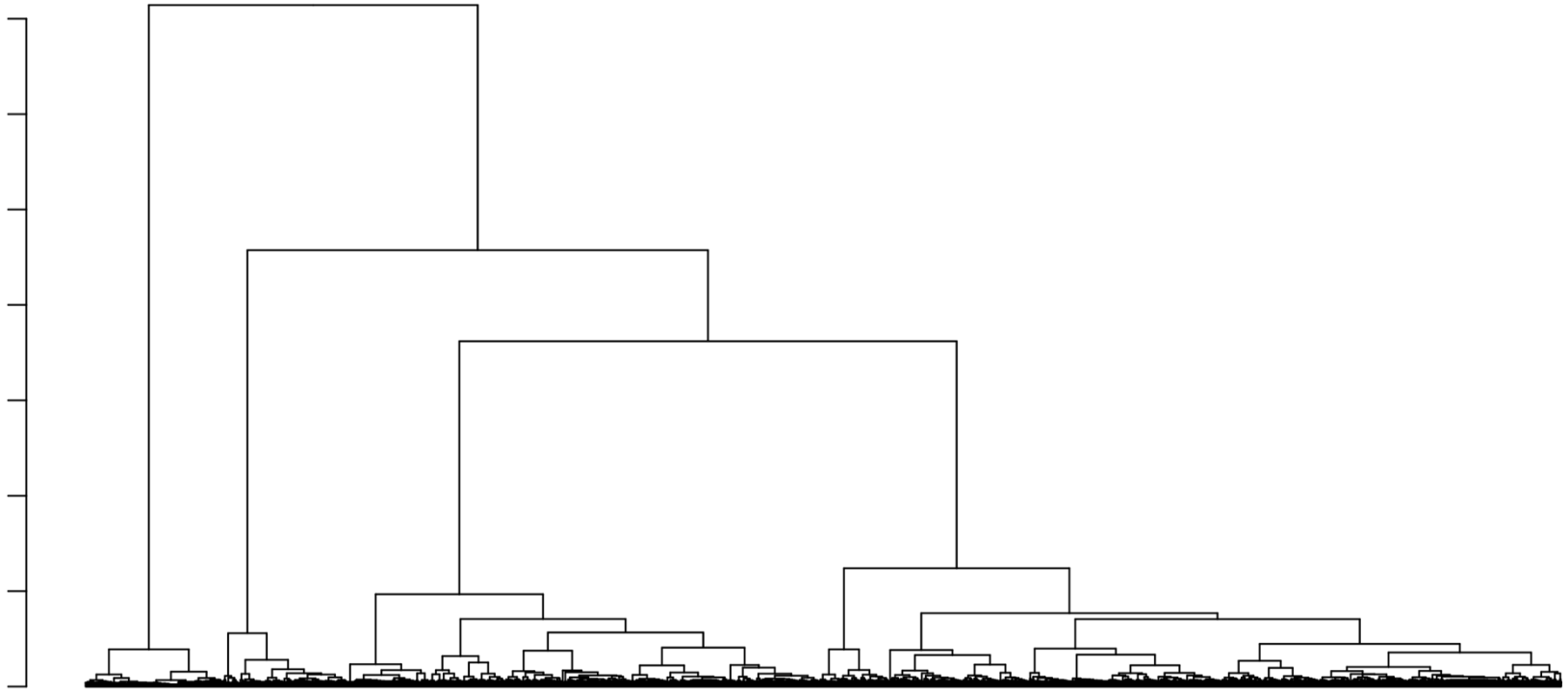
Facets: one small plot per group



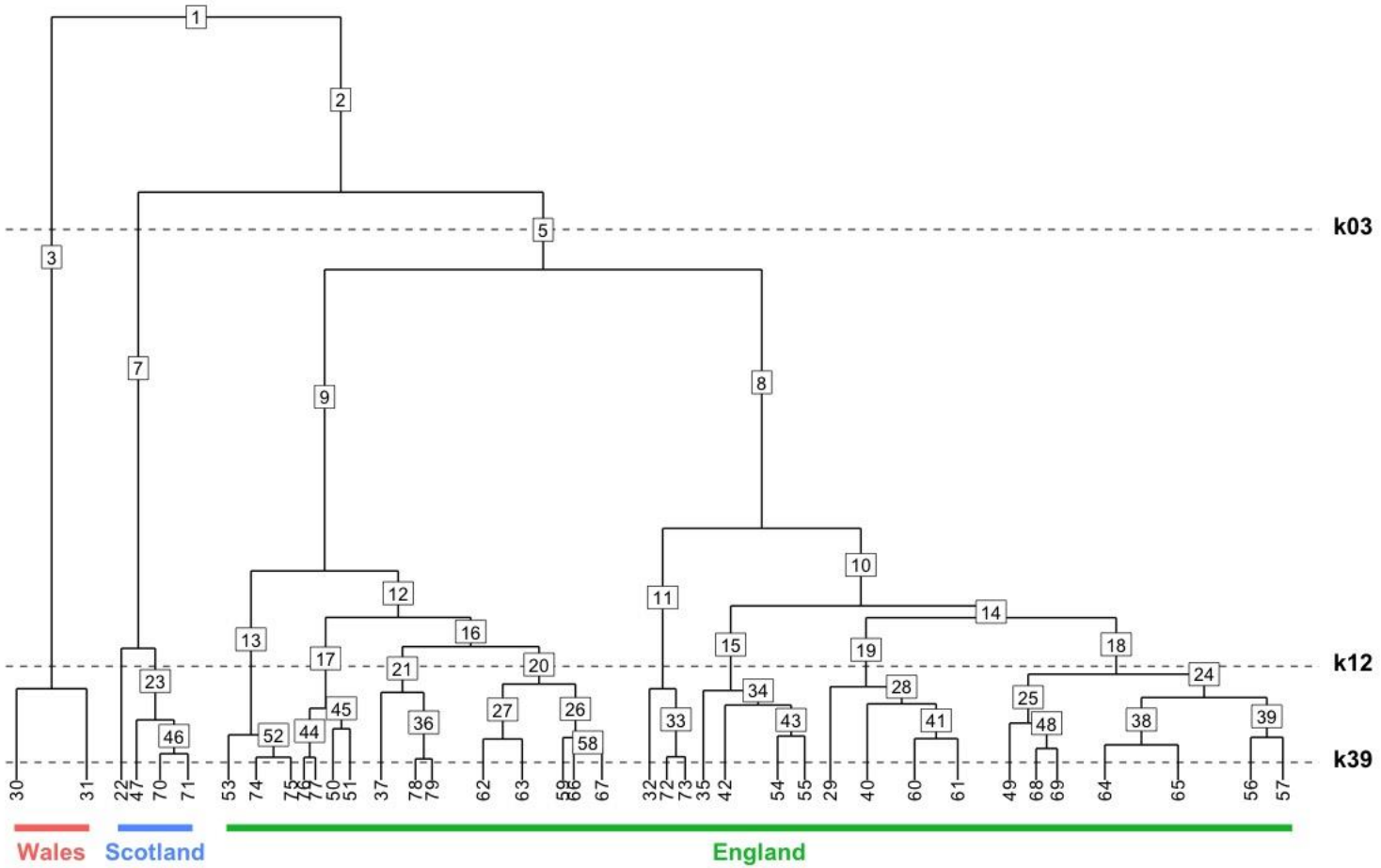
Treemap: area shows each group's share



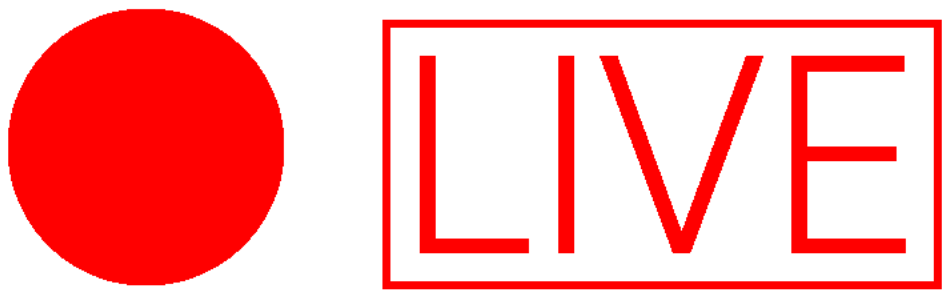
A first plot with the default settings



The same data as a publishable plot



RStudio



Package and code management

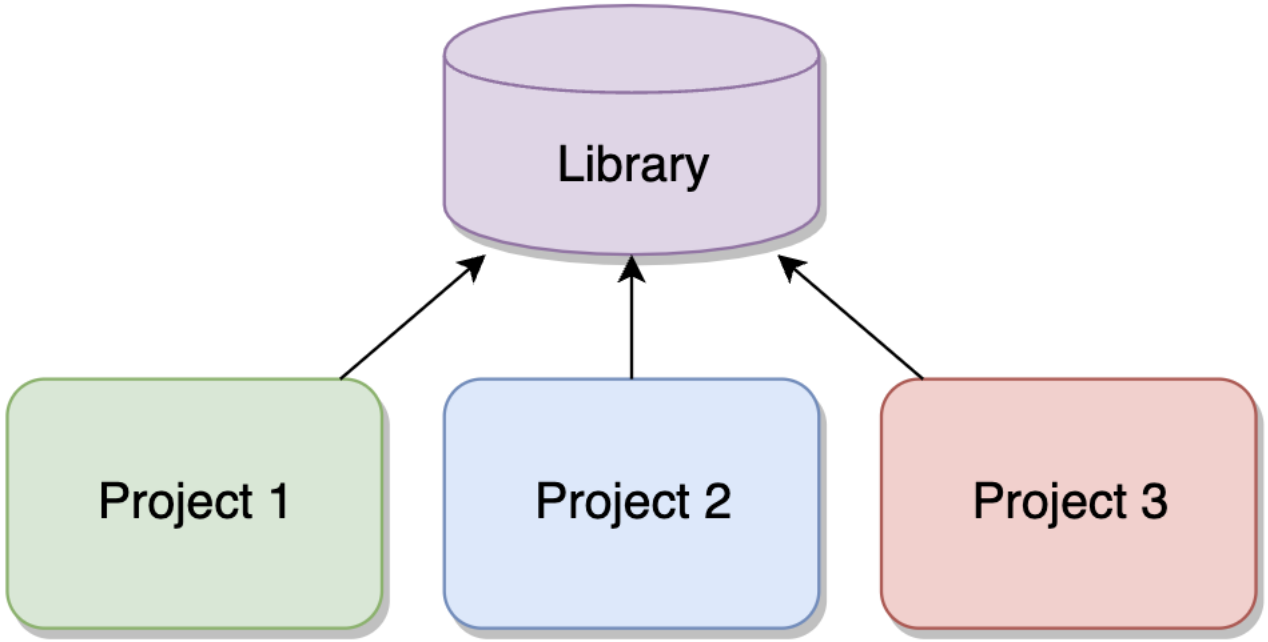
Package management handles dependencies

- Package management is the process of handling the many and varied dependencies and artifacts for your servers, applications, and developers.
- Toolkit used to manage project-local libraries.
- Combination with version management through Git.

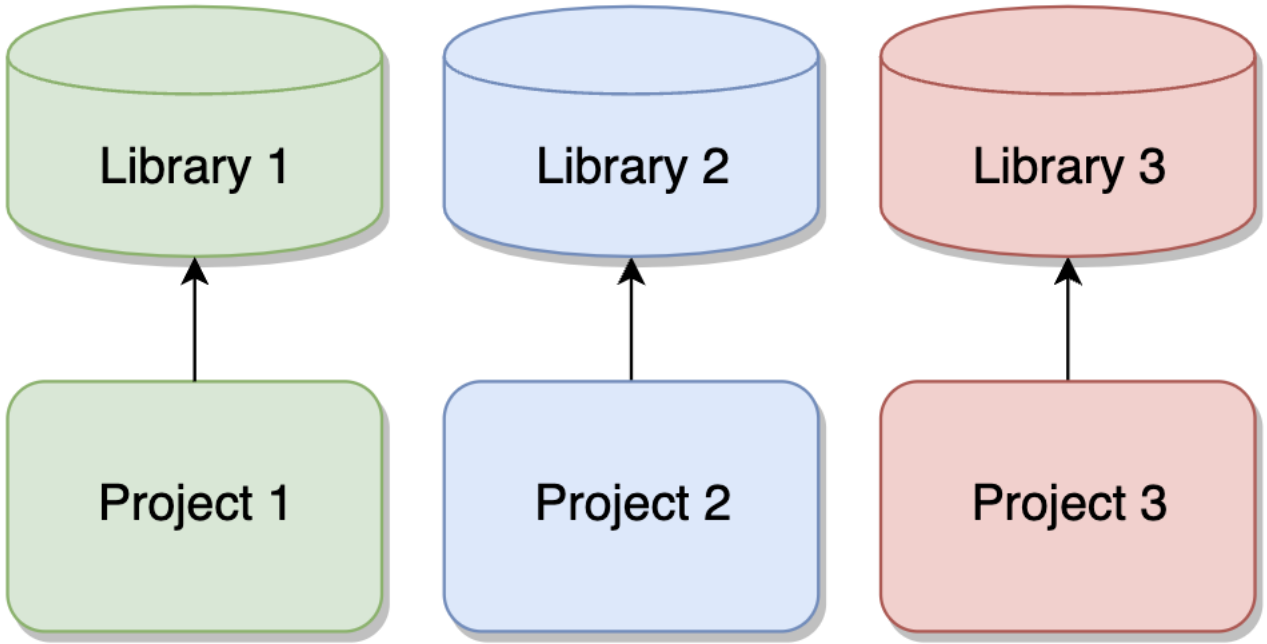




One shared library: one version for every project



Project-local libraries keep projects apart



Version control tracks changes to code

- Version control: The practice of tracking and managing changes to software code.
- Example: Git (GitHub, GitLab).
- “Track changes" on steroids.
- Beyond the scope of this module but you can still set up your own small version control system (“`v0.1`”, “`v0.2`”, “`v1.0`”).

Working with an agentic assistant

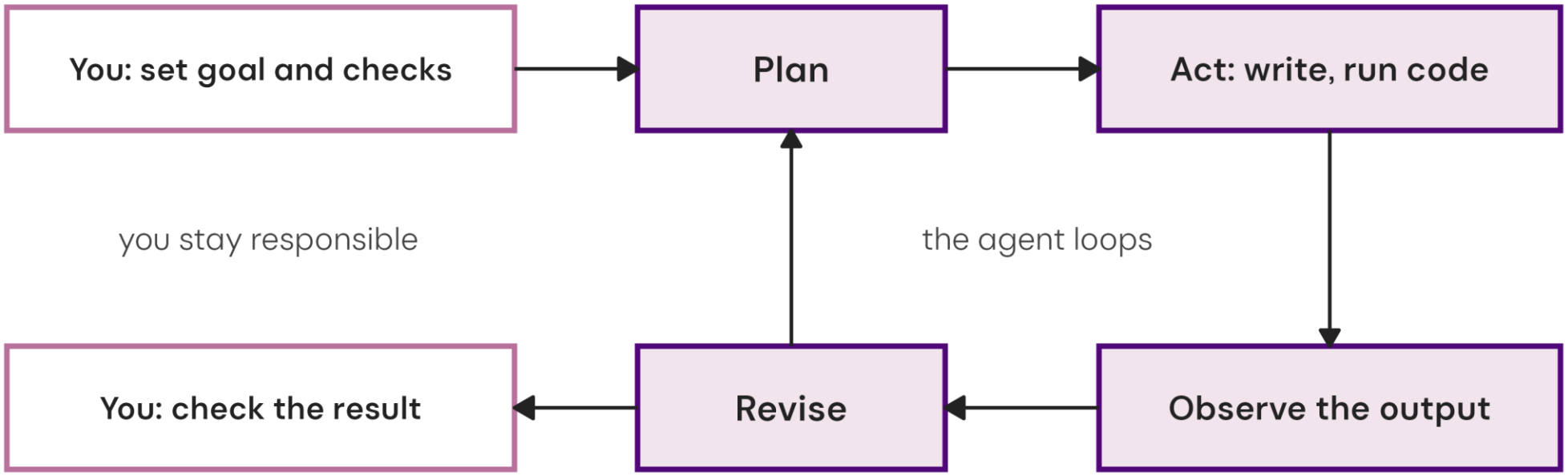
From chat assistant to agent

- A chat assistant answers a prompt. You copy the code, run it and check it.
- An agentic assistant can plan steps, write and edit files, run code, read the output and try again, largely on its own.
- The model underneath still predicts plausible text and code. What is new is the loop around it: tools, files and feedback.
- More autonomy means that less of the work passes in front of you.

Where agents help

- Well-defined, repetitive work: boilerplate, converting or refactoring code, documentation and tests.
- Exploring an unfamiliar library or data set quickly.
- Many small attempts, where each result can be checked automatically.
- Less helpful where the target is unclear or the check is a judgement call: which classification, which neighbourhood definition, which question.

The agent loop, and where you come in



How agents fail silently

- The code runs but the result is wrong: a wrong CRS or unit, a join that adds or drops rows, a default that does not suit the data.
- A failing check is changed instead of the code that fails.
- Changes you did not ask for, made along the way.
- A confident summary that does not match what was done.
- More steps mean more places for an error to hide, and you see the summary, not every step.

Supervising an agent

- State the goal, the data and the checks first: what should the answer look like?
- Work in small steps and read what changed.
- Keep your work under version control, so that any change can be undone.
- Check results against something independent: a total, a known value, a map you can judge.
- Do not give it data or access you would not give a stranger. Stop and take over when you cannot explain a step.

Keep the code, not the chat

- You can only reproduce what is written down in code.
- The same prompt can give a different answer next time.
- Keep a script that runs from top to bottom, and record the package versions.
- Note which tool you used and what you asked it, next to the code.

Check, do not trust: four questions for any result

- What should the answer look like?
- Does it have the right size, unit and number of rows?
- Does it agree with something independent?
- Can I explain every step?

Everything we covered



Choosing a technique: one question per week

Week	Question	Technique
1	How do I make my work repeatable?	Scripts, projects, reproducible workflows
2	What lies near something?	Spatial queries, buffers, overlays
3	Where do points cluster?	Kernel density estimation, DBSCAN
4	Are similar values clustered?	Moran's I and local Moran's I
5	Does a relationship vary by place?	Geographically weighted regression
6	What lies between my measurements?	Raster analysis and interpolation
7	Which areas are alike?	Geodemographic classification
8	How well can people reach services?	Network and accessibility analysis
9	How do I show this clearly?	CRS, classification, bivariate maps
10	How do I check what a tool produces?	Grammar of graphics, AI-assisted workflows

A way of working

- Choose the technique from the question, not the other way round.
- Weeks 1 to 8: do it by hand, so that you know what a correct result looks like.
- Weeks 9 and 10: work with an assistant, and check what it produces.
- Keep the code, record the versions, check the numbers.

Exam

Some pointers for the exam



Some pointers for the exam

- Two-hour timed exam scheduled for TBC
- Two essays (1,000-word limit per question), aim to spend one hour on each.
- Six questions to choose from a range of topics covered in the lectures.
- No questions on coding or programming: focus on theory, underlying principles, methods and applications.
- Q&A Session on TBC.

How to prepare

- Refer to previous exams to get an idea of the types of questions.
- Strategically read the relevant chapters in the textbook (Longley *et al.* 2015) and more theoretical papers.
- Strategically read the more applied papers, make notes of relevant examples or applications of methods.

Example question #1

- Critically reflect on the strengths and weaknesses of Kernel Density Estimation in identifying spatial clusters. In your answer make use of Cheshire and Longley (2012).

Example question #2

- The NHS wants to improve their ambulance response times and they have asked you to assess their current response times in London. Outline the data that you would request, the analytical steps you might take to complete this task and the final maps you could produce in response.

Questions

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Before you go - CMD

- Go to www.menti.com
- Use code: XXXX XXXX

Thank you!

